

1.5 Solving Quadratic Equations

Part 2: Solve Using QF

Lesson Outline:

Part 1: Do It Now - QF Refresher

Part 2: Discriminant review

Part 3: Find exact solutions of a quadratic with 2 roots

Part 4: Solve a quadratic with 1 solution

Part 5: Solve a quadratic with 0 solutions

Part 6: Use the discriminant to determine the number of solutions (x-intercepts) a quadratic has

Part 7: Application

DO IT NOW!

a) Do you remember the quadratic formula?

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

b) Use the quadratic formula to find the x-intercepts of:

$$0 = 2x^2 + 7x - 4$$

Don't forget that to solve a quadratic, it must be set equal to zero because at an x-intercept, the y-coordinate will be zero.

$$x = \frac{-7 \pm \sqrt{(-7)^2 - 4(2)(-4)}}{2(2)}$$

$$x = \frac{-7 \pm \sqrt{49 + 32}}{4}$$

$$x = \frac{-7 \pm \sqrt{81}}{4}$$

$$x = \frac{-7 + 9}{4} \quad \text{or} \quad x = \frac{-7 - 9}{4}$$
$$= \frac{1}{2} \quad \quad \quad = -4$$

Part 2: Discriminant Review

Do all parabolas have two x-intercepts?

NO

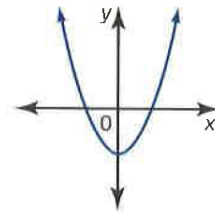
What are the three different scenarios?

0, 1, or 2 solutions

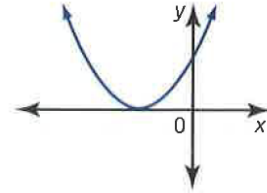
The way to determine how many x-intercepts a parabola might have is by evaluating the $b^2 - 4ac$ part of the quadratic formula (called the "**discriminant**")

Discriminant: the value under the square root

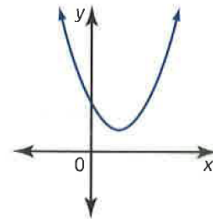
If $b^2 - 4ac > 0$, there are two solutions



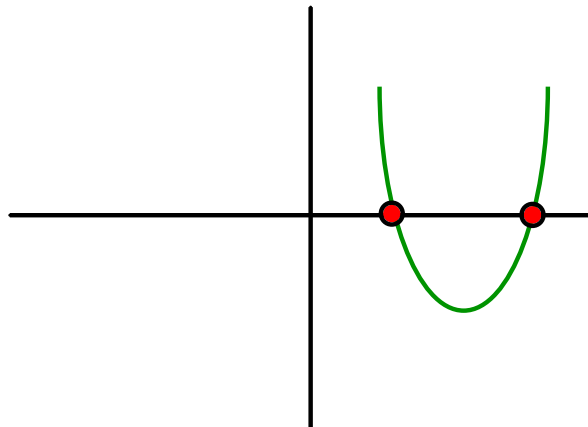
If $b^2 - 4ac = 0$, there is one solution



If $b^2 - 4ac < 0$, there are no solutions



Part 3: Solve a Quadratic With 2 Roots



Objective: Determine the roots of a quadratic using the quadratic formula and leave as EXACT answers

Exact answer: as a radical or fraction. Exact answers do not have decimals.

Example 1: Find the exact solutions of

$$3x^2 - 10x + 5 = 0$$

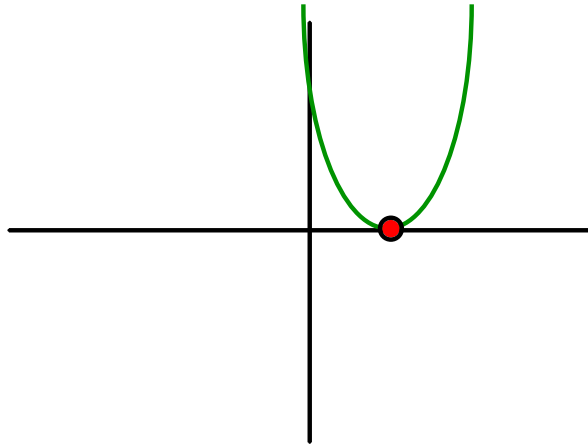
$$\begin{aligned}x &= \frac{10 \pm \sqrt{(-10)^2 - 4(3)(5)}}{2(3)} \\&= \frac{10 \pm \sqrt{40}}{6} \quad (\text{simplify the radical}) \\&= \frac{10 \pm \sqrt{4}(\sqrt{10})}{6} \\&= \frac{10 \pm 2\sqrt{10}}{6} \quad (\text{common factor}) \\&= \frac{\cancel{10} (5 \pm \sqrt{10})}{\cancel{3} 3} \quad (\text{reduce}) \\&= \frac{5 \pm \sqrt{10}}{3} \\x &= \frac{5 + \sqrt{10}}{3} \quad \text{or} \quad x = \frac{5 - \sqrt{10}}{3}\end{aligned}$$

Example 2: Find the exact solutions of

$$-2x^2 + 8x - 5 = 0$$

$$\begin{aligned}x &= \frac{-8 \pm \sqrt{(8)^2 - 4(-2)(-5)}}{2(-2)} \\&= \frac{-8 \pm \sqrt{24}}{-4} \quad (\text{simplify the radical}) \\&= \frac{-8 \pm \sqrt{4}(\sqrt{6})}{-4} \\&= \frac{-8 \pm 2\sqrt{6}}{-4} \quad (\text{common factor}) \\&= \frac{\cancel{-2} (4 \pm \sqrt{6})}{\cancel{2} -4} \quad (\text{reduce}) \\&= \frac{4 \pm \sqrt{6}}{2} \\x &= \frac{4 - \sqrt{6}}{2} \quad \text{or} \quad x = \frac{4 + \sqrt{6}}{2}\end{aligned}$$

Part 4: Solving a Quadratic With 1 Root



Note: when a quadratic only has 1 solution, the x-intercept is also the vertex

Example 3: Find the exact roots of

$$4x^2 + 24x + 36 = 0$$

$$4(x^2 + 6x + 9) = 0 \quad (\text{common factor})$$

$$x^2 + 6x + 9 = 0$$

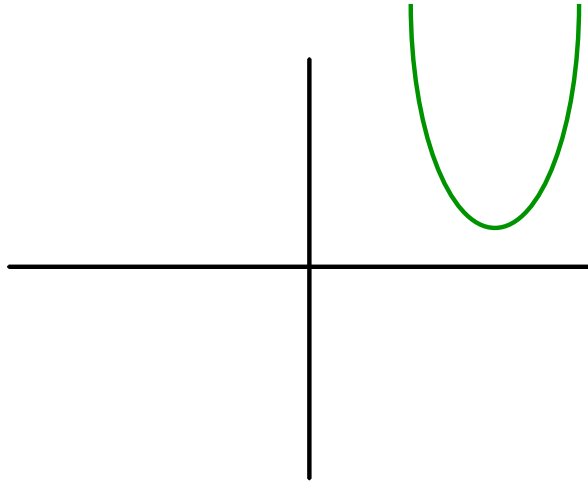
$$x = \frac{-6 \pm \sqrt{6^2 - 4(1)(9)}}{2(1)}$$

$$x = \frac{-6 \pm \sqrt{0}}{2}$$

$$x = \frac{-6}{2}$$

$$x = -3$$

Part 5: Solving Quadratics With 0 Roots



2 Scenarios causing 0 roots:

- i) vertex is above the x-axis and opens up
- ii) vertex is below the x-axis and opens down

Example 4: Find the x-intercepts of

$$8x^2 - 11x + 5 = 0$$

$$x = \frac{11 \pm \sqrt{(-11)^2 - 4(8)(5)}}{2(8)}$$

$$x = \frac{11 \pm \sqrt{-39}}{16}$$

& no solutions

Part 6: Use the Discriminant to Determine the Number of Roots

Example 5: For each of the following quadratics, use the discriminant to state the number of roots it will have.

$$\begin{aligned} \text{a) } 2x^2 + 5x - 5 &= 0 & b^2 - 4ac &= 5^2 - 4(2)(-5) \\ & & &= 25 + 40 \\ & & &= 65 \\ & & 65 &> 0 \text{ ; 2 solutions} \end{aligned}$$

$$\begin{aligned} \text{b) } 3x^2 - 7x + 5 &= 0 & b^2 - 4ac &= (-7)^2 - 4(3)(5) \\ & & &= 49 - 60 \\ & & &= -11 \\ & & -11 &< 0 \text{ ; no solutions} \end{aligned}$$

$$\begin{aligned} \text{c) } -4x^2 + 12x - 9 &= 0 & b^2 - 4ac &= (12)^2 - 4(-4)(-9) \\ & & &= 144 - 144 \\ & & &= 0 \\ & & 0 &= 0 \text{ ; 1 solution} \end{aligned}$$

Part 7: Application

Example 6: A ball is thrown and the equation below model it's path:

$$h = -0.25d^2 + 2d + 1.5$$

'h' is the height in meters above the ground and 'd' is the horizontal distance in meters from the person who threw the ball.

a) At what height was the ball thrown from? *same for 'h' when d=0*

$$\begin{aligned} h &= -0.25(0)^2 + 2(0) + 1.5 \\ &= 1.5 \text{ meters} \end{aligned}$$

b) How far has the ball travelled horizontally when it lands on the ground?

$$0 = -0.25d^2 + 2d + 1.5$$

$$0 = -0.25(d^2 - 8d - 6)$$

$$0 = d^2 - 8d - 6$$

$$d = \frac{8 \pm \sqrt{(-8)^2 - 4(1)(-6)}}{2(1)}$$

$$= \frac{8 \pm \sqrt{88}}{2}$$

$$\begin{aligned} d &= \frac{8 + \sqrt{88}}{2} \\ &= 8.7 \text{ meters} \end{aligned}$$

or

$$\begin{aligned} d &= \frac{8 - \sqrt{88}}{2} \\ &= -0.7 \text{ meters} \end{aligned}$$

reject